

ESP32-2432S024 | I2C on CN1 (IO22 / IO21) | onboard amp, no external amp | SW1 auto-start mod | rev D - all bench-verified

Set the DIP switches to I2C. Address 0x24.
VCC 3.3V GND SDA SCL - all from CN1.
Read once at boot / cart insert only.
NTAG21x UID + NDEF text record -> ROM filename.
Remove the DMG RF shield or the read will fail.
Spiral face toward the cart; ferrite behind it.

ONE CONNECTOR = WHOLE I2C BUS MCP23017 @ 0x20 PNP

A0/A1/A2 -> GND (address 0x20)
RESET -> 3.3V (never leave it floating)
 VDD 3.3V VSS GND SDA SCL - all from CN1.
 GPPU = 0xFF (internal pull-ups), buttons active LOW.
 INT unused - poll once per frame.
 Port B entirely free.

A7-A0

x24

3.3V red

SCL IO21 black

SDA IO22 yellow

GND green

Bottom 8 header pins run in order to A7 -> A0.

UP -> A7 START -> A3
DOWN -> A6 SELECT -> A2
LEFT -> A5 A -> A1
RIGHT -> A4 B -> A0

GND bridged separately from the PCB header to module GND.

X, Y, L, R pads unused - a DMG shell has no buttons for them.

Debounce ~8 ms in software.

Combos:

- Start+Select menu
- Select+Up/Down volume
- Select+Left/Right bright

Mask the D-pad from the emulator while Select is held.

SPDT slide switch:
 centre -> CYD BAT +
 ON -> battery +
 OFF -> GND

CYD BAT - straight to battery -.
 All grounds common.

Grounding CYD BAT+ in the OFF position
 dumps the power IC's input caps, so it
 restarts instantly instead of waiting ~15 s.

**Verify break-before-make, then the
 truth table with the battery UNPLUGGED:**

bat+ to GND	open in BOTH
CYD+ to GND	open ON / short OFF
bat+ to CYD+	short ON / open OFF

**Glue the centre pair so they cannot be
 swapped. Mark the ON end of the switch.**

The board's power IC is an FM5324GA (IP5306 clone). It needs a 50 ms - 2 s LOW pulse on KEY to start. A wire bridge holds KEY low forever and never starts.

ACROSS SW1's PADS:
10 uF electrolytic, + to KEY pad
220 k in parallel

Identify the pads with a meter:
three read common = GND. The odd one out is KEY. Stripe of the cap goes to any of the three ground pads.

Bend both leads flat, trim to ~3 mm, lay the cap against the board. Encapsulate with hot glue - it is the strain relief.

Do NOT solder to the metal housing.
It is floating and connects nothing.

Board photo from the vendor listing, annotation arrows removed. Pin assignment



- 1. SCL is IO21, not IO27.**
IO27 drives the backlight FET (Q2) on this board - confirmed against the factory schematic. IO21 is free.
- 2. Wire colours are NOT conventional.**
On these pigtails BLACK is SCL and GREEN is ground. Go by position on CN1, never by colour habit.
- 3. SW1 needs the cap mod to auto-start.**
See the box at left. A wire bridge cannot work.
- 4. The OFF position must ground CYD BAT+.**
Without it the power IC holds charge and refuses to restart for ~15 s.
- 5. Audio needs no modification.**

confirmed against the factory schematic (ESP32-2432S024-V1.0). Schematic reference - not to scale.

8 ohm, ~36 mm.
Reuse the original DMG driver. 2-pin JST.

**Onboard amp tested OK.
No external amp needed.**

IO4 is NOT an amp enable - the vendor datasheet is wrong. There is no hardware mute, so volume OFF holds the DAC at 128, not 0.

Gain network already sane:
R7 47k / R8 22k / R9 68k.

BLACK IS NOT GROUND HERE

-  GREEN GND CN1 pin 1
-  YELLOW SDA IO22 CN1 pin 2
-  BLACK SCL IO21 CN1 pin 3
-  RED 3.3V CN1 pin 4
-  Battery + via switch
-  Speaker +/-
-  Buttons A7-A0

STILL TO VERIFY

IO34 divider ratio, against a meter, for the low-battery cutoff.

Lowest-power state must stay above ~50 mA or the power IC shuts down after 32 s. No deep sleep, floor the brightness, keep 240 MHz.
